Sharing economy in geotag: what are the travelers' goals sharing their locations by using geotags in social network sites during the tour?

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Abstract

Purpose – *The purpose of this paper is to investigate what triggers tourist's use of geotag as an information sharing tool in social media.*

Design/methodology/approach – *This study divided tourists' goals into task-involved goals (enjoyment of geotags and altruism) and ego-involved goals (anticipated extrinsic reward and desire for attention), and then examined the influences of these goals on their geotag satisfaction and information-sharing behavior by using PLS-Graph 3.0.*

Findings – *Whereas the anticipated extrinsic rewards, altruism, and enjoyment of geotags were found to influence their geotag satisfaction, desire for attention was not. Enjoyment of geotags was found to be the strongest predictor of tourists' geotag satisfaction, which in turn affected their information-sharing behaviors. Based on these findings, the authors present theoretical and practical implications with suggestions for future research.*

Research limitations/implications – *Geotag services are not identical in all social media, so study participants might have perceived the characteristics of geotags differently depending on which social media they use.*

Originality/value – *The enjoyment of geotags, altruism, and anticipated reward were found to influence geotag satisfaction; however, desire for attention was not. The results imply that enjoyment of geotags and anticipated reward strongly predict geotag satisfaction.*

Keywords *Geotag, Social networking sites, Goal theory, Travel information sharing*

Paper type *Research paper*

1. Introduction

Because of the explosive growth of smartphones, which are usually equipped with a camera and a global positioning system service, social media users can easily post their location information and photos with geotags (Okuyama and Yanai, 2013). Every year, Instagram, a popular social media platform, reveals the most geotagged places. The ten most geotagged destinations posted to Instagram in 2014 were world-famous tourist destinations like amusement parks (e.g. Disneyland and Gorky Park), shopping malls (e.g. Siam Paragon and Dubai Mall), baseball and basketball stadiums (e.g. Dodger Stadium, Yankee Stadium, and Madison Square Garden), famous squares (e.g. Times Square and Red Square), and museums (e.g. Louvre) (Wong, 2014). In other words, social media users tend to geotag places where they are enjoying special touristic experiences, not their homes and workplaces.

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Since geotag service is additional location-based tagging service provided by social media, this phenomenon can be understood in the context of tourists' goals for using social media. Many studies have already investigated the factors affecting tourists' use of social media, finding that tourists' desire to enhance their reputations (e.g. Guo *et al.*, 2007; Stasiak, 2013); attract others' attention (e.g. Kim and Tussyadiah, 2013; Stasiak, 2013); help other people (such as potential tourists) (e.g. Koo *et al.*, 2015; Xiang and Gretzel, 2010; Yoo and Gretzel, 2010); or simply seek pleasure (e.g. Kang and Schuett, 2013) are predictors of their use of social media. Therefore, geotagging can be regarded as a useful information-sharing practice that helps tourists achieve their goals in social media.

Although geotagging plays an important role in tourists' information sharing, academic research has only begun to investigate geotags, and most researchers have tried to analyze and track tourists' locations and routes by using tourists' geotagged postings (e.g. Okuyama and Yanai, 2013; Zheng *et al.*, 2011). As a result, little attention has been paid to the variables that influence and are influenced by tourists' satisfaction with geotags. As geotags are a novel information-sharing tool, the relationship between tourists' goals and satisfaction with geotags and the impact of such satisfaction on their travel information-sharing behavior is worth investigating.

Furthermore, the geotag is relevant to the smart city, an "urban environment which, supported by pervasive Information Communication Technology (ICT) systems, is able to offer advanced and innovative services to citizens in order to improve the overall quality of their life" (Piro *et al.*, 2014, p. 169). The smart city strongly promotes the formulation of smart tourism destinations by providing both cities and tourists with access to value-added services such as real-time information (Buhalis and Amaranggana, 2013). Geotags are also real-time location-based information provided by tourists and are important in disseminating information about the city. This kind of information is a key factor for a successful smart city (Gretzel *et al.*, 2015). In this vein, study on geotags can be considered as meaningful attempts from the perspective of smart tourism.

Therefore, the aim of this study is twofold. First, by adopting goal theory (House, 1971), we categorize tourists' goals as task- or ego-involved. Second, we investigate the effect of the two kinds of goals on geotag satisfaction and travel information-sharing behavior.

The rest of this study organized as follows. Section 2 presents the literature on geotag, goal theory and process theory, followed by the research model and corresponding hypotheses in Section 3. Section 4 describes the measures and data collection procedures use in the research. The analysis procedures and results are detailed in Section 5. Section 6 offers conclusions, implications and limitations.

2. Theoretical background

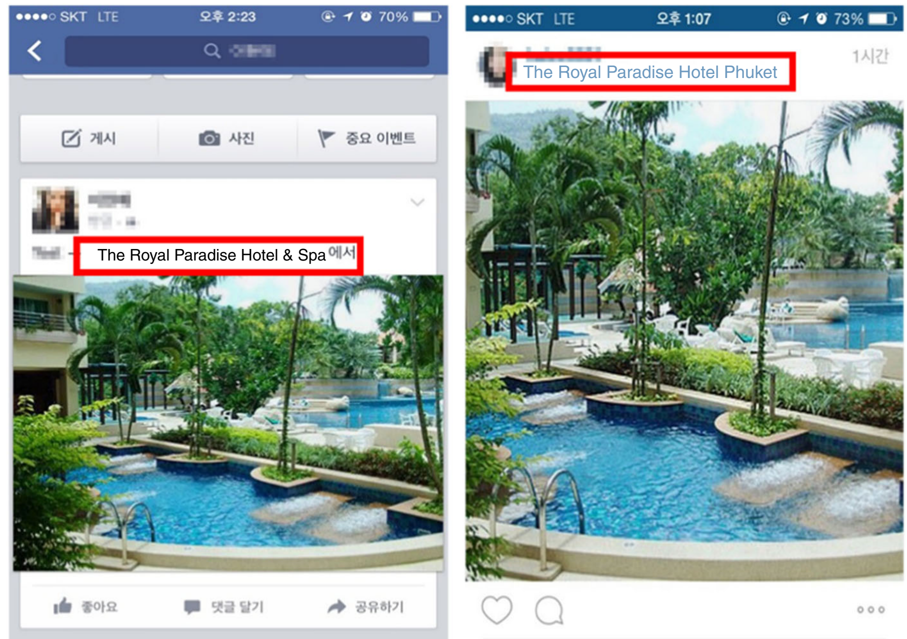
2.1 Geotags

Social media with location-based technology encourage the consumption of places (Tussyadiah, 2012). Geotagging is popular among users of location-based services (Buczkowski, 2012). Geotags are geographical identification metadata added to media to identify one's location for other people (Gikas and Grant, 2013) and location-based tagging services provided by social media such as Facebook, Flickr, Foursquare, Instagram, and Twitter. With the explosive growth of social media, the majority of tourists have actively used geotags when posting their travel experience and information on social media (Okuyama and Yanai, 2013).

Geotags offer a feature of visualizing some contents in social media (Buczkowski, 2012). For instance, they show the precise name of the location, such as luxurious and prestigious hotels or resorts, and appear different from general text: blue in Instagram and bold in Facebook (see Figure 1). This geotag feature is able to fully satisfy tourists' desires to identify their locations to other people in order to show off their touristic experience (Stasiak, 2013) and help other people by providing useful and obvious touristic information (Yoo and Gretzel, 2010).

Therefore, we argue that geotags can be regarded as a useful information-sharing tool for tourists that can also contribute to the effective management of tourism businesses. However, little

Figure 1 Snapshots of geotags



Notes: (Left: Facebook, Right: Instagram)

attention has been paid to geotags, and only a small number of studies have identified regions of attraction and analyzed tourists' travel paths using geotags (e.g. Okuyama and Yanai, 2013; Zheng *et al.*, 2011).

2.2 Goal theory

Tourists use social media for many reasons. According to some studies, tourists post their experiences and information in social venues to enhance their reputation (e.g. Guo *et al.*, 2007; Stasiak, 2013), attract others' attention (e.g. Kim and Tussyadiah, 2013; Stasiak, 2013), help other people (such as potential tourists) (e.g. Koo *et al.*, 2015; Xiang and Gretzel, 2010; Yoo and Gretzel, 2010), and seek pleasure (e.g. Kang and Schuett, 2013). The present study examined these goals by using goal theory as its theoretical framework.

Goal theory postulates that people's behavior and performance can be influenced by their goals (House, 1971). Nicholls and Attwell (1990) identified two types of goals: task- and ego-involved. The former emphasizes "gaining insight or skill" and developing competence; the latter focusses on "establishing one's ability as superior to that of others" (Nicholls and Attwell, 1990, p. 110). Therefore, people with a task-involved perform the behavior in order to enhance their ability or seek pleasure, making them intrinsically motivated; in contrast, people with an ego-involved goal tend to strive for social recognition or higher social status, making them extrinsically motivated (Deci and Ryan, 1985; Duda, 1996).

Goal theory has been adopted by researchers with an interest in examining (physical) education (e.g. Covington, 2000; Nicholls *et al.*, 1985; Xiang *et al.*, 2004) and sports (e.g. Duda *et al.*, 1995; Ntoumanis, 2001); however, a few researchers have applied goal theory to the tourism context. Kassim *et al.* (2014) investigated the impact of task- and ego-oriented goals of golf tourists on flow experience; their results revealed a significant relationship between ego orientation and flow experience.

In the context of tourists' use of geotags, task-involved tourists presumably seek pleasure or to provide touristic information by posting their location; ego-involved tourists prefer to show off the luxury hotel or resort where they are staying in order to enhance their social status or attract attention. For instance, tourists can achieve the task-involved goals of deriving pleasure from helping others by sharing more useful and reliable information and experience with geotags, whereas tourists can achieve ego-involved goals by receiving social rewards such as reputation enhancement and attention from others by identifying the luxury hotels, resorts, and tourism sites that they have visited.

Further, when these goals are achieved by using geotags, they increase the possibility of positive evaluation and intention to use geotag again. We regarded enjoyment of geotags and altruism as task-involved goals; anticipated extrinsic rewards and privacy unconcern are ego-involved goals. We investigated whether these goals can increase users' satisfaction and information-sharing behavior when using geotags.

2.3 Process theory

Process theory explains the temporal order and sequence between input (independent variables) and output (dependent variables), and places emphasis on the observed sequence of events which enable results to happen (Van de Ven and Huber, 1990). Process theory has been used in behavioral research (e.g. Jung *et al.*, 2015), and satisfaction has been used as a process state (e.g. Burke *et al.*, 2001; Jung *et al.*, 2015). Satisfaction is defined here as "a person's critical evaluation of several aspects of perceptions, attitudes and personal value" (Demers *et al.*, 2002, p. 101), and has been investigated as a mediator between positive perceptions of technologies or online platforms and intention to use them (Chung *et al.*, 2015). For instance, Jung *et al.* (2015) used process theory to explain the sequences from the qualities of augmented reality (AR) (input) to intention to recommendation (output), and used AR satisfaction as an intervening construct (process).

If tourists' task- or ego-involved goals were reached by geotagging, tourists can positively evaluate geotags and ultimately intend to use them. Therefore, this study has three phases of tourists' task- and ego-involved goals (input), geotag satisfaction (process), and information sharing behavior (output).

3. Hypothesis development

In order to investigate the impacts of task-involved goals and ego-involved goals on geotag satisfaction and travel information-sharing behavior, we propose the conceptual model shown in Figure 2. We propose five hypotheses and, in this section, present the theoretical background for each hypothesis.

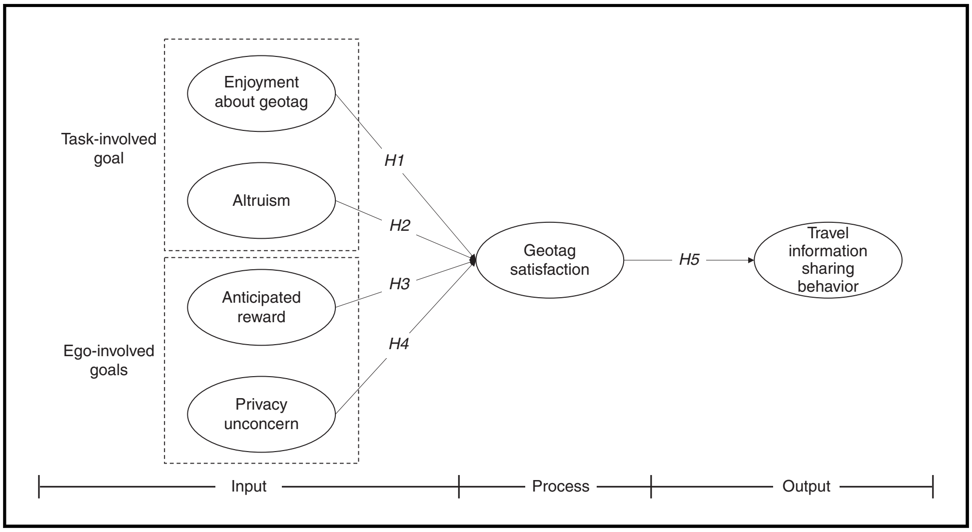
3.1 Task-involved goal and geotag satisfaction

According to Deci and Ryan (1985), a task-involved goal stimulates intrinsic motivation. Intrinsically motivated people tend to engage in a behavior in order to derive pleasure, interest, enjoyment, and a sense of challenge from the behavior (Moos and Marroquin, 2010), rather than to gain an extrinsic reward. In terms of social media, people share their touristic information because they enjoy helping potential tourists (Xiang and Gretzel, 2010) or derive pleasure from the act itself (Kang and Schuett, 2013).

According to Foux (2006) and Kang and Schuett (2013), user-generated touristic information or experiences are more reliable than information from tourism organizations or companies. Social media users provide other social media users with information about their location by using geotags (Gikas and Grant, 2013). Thus, it can be assumed that people consider geotags as useful information and experience-sharing tools for helping others and that they are satisfied with them.

Geotags are different from general text (e.g. blue in Instagram, bold in Facebook). Previous studies have postulated that design characteristics or visual aesthetics of mobile applications influence people's perceived enjoyment (e.g. Cyr *et al.*, 2006). Perceived enjoyment is the extent to which someone enjoys using a computer (Van der Heijden, 2003), and is one of the strongest

Figure 2 Research model



predictors of positive evaluation about the cutting-edge technologies. The perceived enjoyment of a geotag can be defined as the extent to which geotagging behavior is perceived as enjoyable, and it can influence satisfaction with them. Thus, the following hypotheses are proposed:

H1. Enjoyment of geotags has a positive impact on geotag satisfaction.

H2. Altruism has a positive impact on geotag satisfaction.

3.2 Ego-involved goal and geotag satisfaction

According to Skinner (1938), rewards can elicit behavior. Although numerous studies have focussed on the relationship between anticipated economic rewards, such as financial benefits and behavior (e.g. Yin *et al.*, 2014), in social media, social rewards such as enhancing reputation (Bitzer *et al.*, 2007) strongly motivate behavior. Social media users tend to post their touristic experiences or information in order to gain a reputation within their virtual community (Stasiak, 2013).

As previously stated, people with ego-involved goals tend to focus on their superiority over others (Nicholls and Attwell, 1990) and engage in behavior to obtain a goal-unrelated achievement (Deci, 1975), such as social reward and attention. Social media users prefer to post their touristic experience and information on social media on the spot in order to enhance their social status by showing off their knowledge, economic ability, and free time (Stasiak, 2013). Geotags display the name of a location, such as luxury hotels or resorts, making geotags a helpful way for identifying tourists' locations to other people (Okuyama and Yanai, 2013; Zheng *et al.*, 2011). Thus, it can be assumed that people with ego-involved goals are easily satisfied with geotags:

H3. Anticipated reward has a positive impact on geotag satisfaction.

H4. Privacy unconcern has a positive impact on geotag satisfaction.

3.3 Geotag satisfaction and travel information-sharing behavior

Satisfaction has a great influence on future behavioral intentions (Oliver, 1980). According to some literature, satisfaction with social platforms and information-sharing tools has significant influence on the intention to continuously use or reuse it (e.g. Chen, 2007; Khan *et al.*, 2014). The results showed that the effect of satisfaction with online virtual communities on intention to keep using it is stronger than the effect of the social interaction tie (Chen, 2007). Furthermore, Khan *et al.* (2014) found that satisfaction with social network sites has a great impact on intention to use them. Accordingly, people satisfied with geotags are more willing to use them. Using geotags can be regarded as information- and experience-sharing behavior as geotags play a role in identifying and sharing tourists' locations and information about them to other people on social media. Therefore, tourists with a high satisfaction with geotags may have strong motivation to share their travel experiences and information by using geotags. This leads to the following hypothesis:

H5. Geotag satisfaction has a positive impact on travel information-sharing behaviors.

4. Research method

4.1 Measures

Measurement items were adopted from previous literature (e.g. Dinev and Hart, 2006; Kim *et al.*, 2007; Oliver, 1996; Yang, 2013). All items were measured on a seven-point Likert scale ranging from strongly disagree (1) to strongly agree (7). This procedure yielded 21 measurement items. Table I summarizes all 21 measurement items by construct: anticipated extrinsic reward (three items), privacy unconcern (four items), enjoyment of geotags (four items), altruism (three items), satisfaction with geotags (four items), and travel information-sharing behavior (three items). The survey questionnaire was developed in English first and then translated into Korean by people

Table I Reliability and cross-loading

Constructs	*Measurement items*	*Loading*	*t-value*	α	*CR*	*AVE*
Enjoyment about geotag (Kim *et al.*, 2007)	I have fun posting geotagged context or photos about tourism sites	0.902	38.236	0.922	0.945	0.811
	Posting geotagged context or photos about tourism sites provides me with a lot of enjoyment	0.923	53.894			
	I enjoy posting geotagged context or photos about tourism sites	0.877	35.129			
	Posting geotagged context or photos about tourism sites do not bores me	0.899	47.763			
Altruism (Yang, 2013)	If my travel experience is pleasant, I want to help others with my own positive travel experience	0.892	17.519	0.833	0.900	0.751
	It feels good to help others on social media	0.852	21.277			
	If I am so satisfied with the tourism site, I want to help the tourism site to be successful	0.854	17.333			
Anticipated reward (Yang, 2013)	Posting geotagged context or photos about tourism sites shows others that I am a clever tourist	0.796	16.968	0.849	0.908	0.768
	I feel that posting geotagged context or photos about tourism sites improves my status	0.923	53.987			
	I post geotagged context or photos about tourism sites to improve my reputation	0.906	40.775			
Privacy unconcern (Dinev and Hart, 2006)	I am unconcerned that a people can find private information about me through the geotag	0.626	3.511	0.943	0.879	0.650
	I am unconcerned about posting geotagged context or photos about tourism sites	0.949	4.235			
	I am unconcerned that posting geotagged context or photos about tourism sites would involve many unexpected problems	0.840	4.266			
	I am totally unconcerned that posting geotagged context or photos about tourism sites would bring about privacy-related problems	0.775	4.170			
Geotag satisfaction (Oliver, 1996)	Overall, I am satisfied with using geotag	0.918	48.901	0.933	0.953	0.836
	Overall, I am glad to use geotag	0.861	21.229			
	Overall, I feel good to use geotag	0.941	80.711			
	Overall, I feel pleasure with using geotag	0.935	67.616			
Travel information sharing behavior (Yang, 2013, revised)	If I am satisfied with travel experience, I will feel good when I can tell others about my great experience by posting geotagged context or photos about tourism sites	0.967	93.052	0.962	0.976	0.930
	If I am satisfied with travel experience, I intend to share my experience with other people by posting geotagged context or photos about tourism sites	0.970	124.074			
	If I am satisfied with travel experience, I intend to say good things about tourism site by posting geotagged context or photos about tourism sites	0.957	92.904			

Notes: α, Cronbach's α; CR, composite reliability; AVE, Average variance extracted who were proficient in both languages. Researchers fluent in English and Korean and with academic specializations in the area under study compared the translated version with the original. No material discrepancies were found. These processes in the pretest stage showed that all questions were valid and reliable.

4.2 Data collection

The survey was administered to 120 university students as geotags have not yet been commercialized enough to be known among a wide range of users. We assumed that most geotag users were young people. Participants were asked to recall a memory and post their touristic experience and information with geotags on social media before participating in the survey. One incomplete questionnaire was eliminated, leaving 119 questionnaires for analysis. Of these 119 respondents, 80 (67.2 percent) were female, and 39 (32.8 percent) were male; most were in their 20s (117, 98.3 percent) and had never used geotags when posting touristic experiences and information on social media (97, 81.5 percent).

5. Analysis and results

In order to examine the proposed research model, we used a partial least squares (PLS) regression analysis by using PLS-Graph version 3.0, as a PLS regression analysis offers several advantages, including a small sample size, and require few assumptions about measurement scale and normal distribution (Ahuja and Thatcher, 2005).

5.1 Measurement model

The conceptual model (Figure 2) was estimated using structural equation modeling. Before assessing the hypothesized relationships, confirmatory factor analysis was conducted by checking the reliability and validity of each item. We estimated Cronbach's α, composite reliability, and average variance extracted (AVE) of each construct in order to check reliability and validity. As shown in Table I, the results showed that the reliability of each item, composite reliability (greater than 0.7), Cronbach's α (greater than 0.7), and AVE (greater than 0.5) satisfied the requirements. Thus, the results established that the items of the present study demonstrated reliability and validity.

The discriminant validity of the measurement model was checked by comparing the square root of the AVE for each construct with the correlations between that and other constructs. If the square root of the AVE exceeded the correlations between the construct and another construct, it indicated discrimination. As shown in Table II, the results indicated that the square root of the AVE for each construct exceeded the correlations between that construct and other construct; thus, the discriminant validity of the instrument was established.

Table II Correlation and discrimination validity

	Mean	*SE*	*(1)*	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*
(1) Enjoyment about geotag	4.574	1.324	*0.901*					
(2) Altruism	5.160	1.201	0.475**	*0.867*				
(3) Anticipated reward	3.734	1.349	0.545**	0.350**	*0.876*			
(4) Privacy unconcern	3.515	1.599	0.007	−0.067	−0.165	*0.806*		
(5) Geotag satisfaction	4.634	1.072	0.649**	0.468**	0.541**	0.018	*0.914*	
(6) Travel information sharing behavior	5.359	1.405	0.488**	0.696**	0.217*	−0.099	0.560**	*0.964*

Notes: SE, Standard error. Diagonal elements in the "correlation of constructs" matrix are the square root of the average variance extracted (AVE). For adequate discriminant validity, the diagonal elements should be greater than the corresponding off-diagonal elements. *$p < 0.05$; **$p < 0.01$

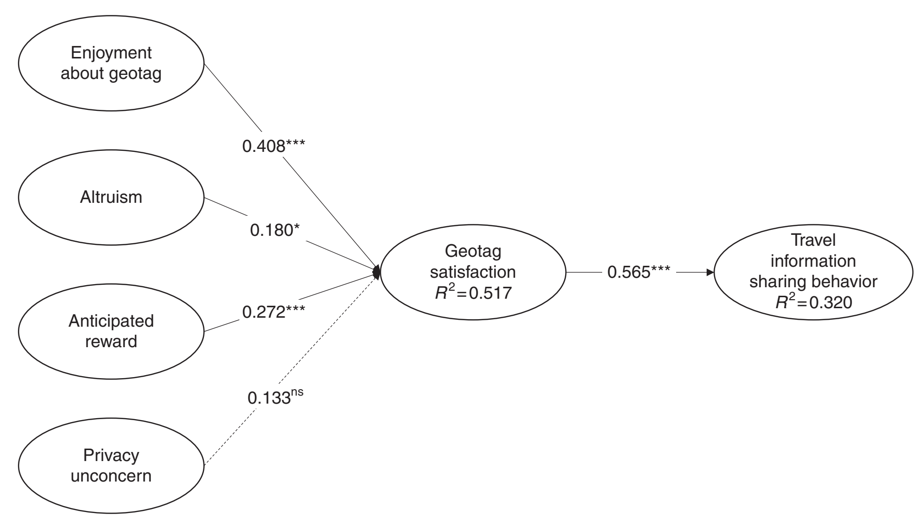
5.2 Structural model

Structural equation modeling was conducted to assess the validity of the proposed research model. The size of the bootstrapping sample used in the PLS analyses was 500. Figure 3 displays the outcome of the structural model, and Table III shows the results of the hypothesis testing. *H1-H3* postulating the impact of enjoyment of geotags ($\beta = 0.408$, $t = 4.350$), altruism ($\beta = 0.180$, $t = 2.154$), and anticipated reward of geotag satisfaction ($\beta = 0.272$, $t = 2.555$) were supported. In particular, enjoyment of geotags was found to be the strongest predictor of geotag satisfaction. *H4* postulating a positive relationship between privacy unconcern and geotag satisfaction was rejected ($\beta = 0.133$, ns), and enjoyment of geotag was found to be the strongest predictor of geotag satisfaction. Finally, the result of the test of *H5* indicated that geotag satisfaction has a significant influence on travel information-sharing behaviors ($\beta = 0.565$, $t = 4.921$). Thus, *H5* was supported.

6. Conclusion

Geotags allow users to reveal their location to other people on social media and can, thus, be both information-sharing and marketing tools. In this vein, the present study investigated the factors influencing travel information-sharing behavior with geotags. Using goal theory, we categorized tourists' goals of using geotags as task-involved goals (e.g. enjoyment of geotags and altruism) and ego-involved goals (e.g. anticipated reward and privacy unconcern) and then tested the impact of these goals on satisfaction with geotags and with travel information-sharing behavior.

Figure 3 Results of the structural equation modeling analysis



Notes: ns = non-significant. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table III Standardized structural estimates and tests of the hypotheses

Hypotheses	*Path*	*Estimates*	*t-value*	*Results*
H1	Enjoyment about geotag → geotag satisfaction	0.408	4.350	Supported
H2	Altruism → geotag satisfaction	0.180	2.154	Supported
H3	Anticipated reward → geotag satisfaction	0.272	2.555	Supported
H4	Privacy unconcern → geotag satisfaction	0.133	1.184	Rejected
H5	Geotag satisfaction → travel information sharing behavior	0.565	4.921	Supported

The enjoyment of geotags, altruism, and anticipated reward were found to influence satisfaction with geotags; however, privacy unconcern was not. These results imply that enjoyment of geotags and anticipated reward strongly predict geotag satisfaction, which was found to affect tourists' information-sharing behavior, thereby implying that tourists who are satisfied with geotags tend to post their touristic experience and information with geotags via social media.

In light of these results, this study has theoretical and practical implications. First, although little attention has been paid to geotags and goal theory in tourism, this study regards geotags as an information-sharing tool for tourists and investigates the factors influencing tourists' satisfaction. Goal theory was originally adopted in (physical) education and sports; we extended the territory of goal theory from the achievement- to sharing-related goals of tourists. We categorized tourists' goals for using geotags and empirically tested the factors influencing geotag satisfaction and touristic information- and experience-sharing behavior.

Second, enjoyment of geotags and anticipated reward were found to be the strongest predictors of satisfaction. These results demonstrate that geotags provide tourists with pleasure by letting them show their cleverness; this finding is similar to those of previous studies in that tourists' perceived enjoyment (e.g. Kang and Schuett, 2013) and anticipated reward (e.g. Guo *et al.*, 2007; Stasiak, 2013) are important in social media. Furthermore, tourism marketers can use these findings in developing marketing strategies; satisfaction with geotags affects touristic experience- and information-sharing behaviors. In this vein, marketers should offer benefits (e.g. discounts and gift certificates) to encourage tourists to use geotags when posting their touristic experience and information to social media during their stay at hotels, resorts, and tourism destinations. Additionally, marketers of hotels, tourism spots, and restaurants should control and manage the geotag page that displays reviews, location, pictures, and other information about the destination.

Finally, since a geotag presents location-based information, it is important in disseminating city-related information, which is critical in the success of the smart city and smart tourism destination (Gretzel *et al.*, 2015). Therefore, this study is worthwhile from the perspective of smart tourism.

However, this study has some limitations. First, geotag services are not identical across social media, so study participants might have perceived the characteristics of geotags differently depending on which social media platform they use. Therefore, future studies should choose specific social media and geotag services. Second, most participants in this study were students; they were young and technologically sophisticated. Thus, it is unclear whether the results generalize to older people with less technological sophistication. Therefore, further studies should assess a wide range of people who post touristic experiences and information with geotags on social media. Finally, although we adopted all of the measurement items from the previous studies, it is possible that conditional sentences of two constructs (altruism and travel information-sharing behavior) led to biased responses. Therefore, future studies should carefully adopt measurement items according to study context or framework.

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